

Avinash Agarwal

Civilization, AI, Healthspan, Space

Mathematics | Engineering | Superintelligence | Systems

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SUMMARY

Inquisitive engineering veteran, with **8 years of industry leading experience at Meta and Microsoft**, passionate about tackling civilizational challenges, especially Healthspan and Space exploration!

EXPERIENCES

Meta

Full-time

August 2025 — Present

Menlo Park, California, USA

- **Architecting Global Identity Systems:** Ensuring high-availability and security for identity systems processing billions of requests across the entire Meta ecosystem.
- **Secured runners-up position in an internal Hackathon**, demonstrating high engineering velocity by rapidly triaging and fixing a high volume of improvements.
- **Contributed to the development of a company-wide automated on-call agent**, significantly reducing manual triage time and improving engineering team productivity."
- **AI for Longevity at Meta Superintelligence Labs:** Spearheading technical advocacy and research to improve AI powered Health Benchmarks.

Microsoft Azure

Full-time

Feb 2022 — July 2025

Redmond, Washington, USA

Azure Resource Graph

- Collaborated with Daniel Hu to migrate a streaming data pipeline (1B+ daily notifications) to its dedicated scale unit — Involved sophisticated design, careful cross-team planning and execution.
- **Root-caused and advocated for increased hash cardinality** across the storage stack, leveraging data-driven demonstrations to significantly improve distribution granularity and resolve partition skewness.
- **High-Performance Serialization:** Evaluated and advocated for the adoption of **System.Text.Json** across the service stack, leveraging its high-performance, low-allocation APIs to achieve significant reductions in CPU overhead and memory pressure during large-scale telemetry processing.
- **Engineered data ingestion components** hosted on the Azure Synapse platform to support dynamic time-window loading for security datasets.
- Optimized complex SQL queries executed on Azure Synapse to efficiently load and merge multiple continuous data streams.
- Profiled and optimized high-contention services eliminating bottlenecks and achieving **nearly 10%** improvement in compute, memory and network efficiency.
- Improved on-call efficiency with multiple improvements to internal monitors, documentations, etc.

Azure Resource Change Analysis

- Delivered General Availability for the Change Analysis back-end by successfully deploying the service to US and China Government Sovereign Clouds, ahead of an important Gartner cloud review.
- Significantly reduced partition skewness in data-ingestion pipelines, achieving **over 30%** improvement in compute, memory and network efficiency.

- Led the architectural evolution, including metadata enhancements and optimizations to the Change Analysis payload with extensive collaboration across teams and disciplines.
- Delivered General Availability for the Change Actor functionality, enabling customers to audit **WHO** initiated a change and **WHICH client** (Portal, CLI, SDK) was used to make these changes.
- Delivered General Availability for the Change Analysis front-end ([live on the Azure portal](#)) by improving [Lighthouse scores](#), automating deployments and tests, completing A11y fixes etc.
- Closely collaborated with the Change Safety team to reference Change Analysis resources.

Azure Global

Full-time

Feb 2022 — July 2025

Hyderabad, Telangana, India

BCDR & Global Traffic Management: Designed and Implemented a mission-critical Business Continuity and Disaster Recovery architecture using [Azure Front Door](#) (already benchmarked by LinkedIn), significantly improving page-load latencies, establishing a resilient global load-balancing layer and hardening the platform's security posture for massive-scale international traffic.

Strategic Architectural Leadership: Formally advocated for [Azure Lighthouse](#) over Azure Managed Applications to resolve cross-tenant visibility constraints; authored deep-dive technical evaluations demonstrating how Lighthouse's automation via Azure Marketplace would have accelerated time to market and eliminated multi-tenant architectural debt for **ALL of Azure Global**.

Scaling for GRAB (Strategic Partnership)

- **Performance Profiling & Load Testing:** Collaborated with [Kinshuk Singh](#) to perform intensive load testing and performance profiling using Apache JMeter, ensuring platform stability for massive-scale deployments across Southeast Asia.
- Identified and contributed fixes to Azure JMeter + ACI Terraform setup (GitHub issue 46), enhancing deployment reliability and aligning with large-scale performance testing initiatives.
- **System Optimization & Throughput:** Optimized core services to achieve a **20x throughput improvement** by studying and refining SQL query execution plans, eliminating redundant network calls, and implementing strategic caching layers.
- **High-Scale Data Migration:** Architected APIs to execute the **migration of 160M+ course-completion records** from AWS to Azure ensuring data integrity during the transition.
- **Technical Leadership & Design:** Spearheaded technical design reviews and PM alignment, advocating for extensible solutions to avoid technical debt from short-term fixes.

UNICEF Learning Passport (powered by Microsoft Community Training)

"Education is the ultimate 'Great Filter' prevention mechanism"

- **Offline Innovation (Hackathon Lead):** Spearheaded a hackathon project to host the entire cloud service locally on Windows machines. The platform was recognized as one of [TIME's Best Inventions of 2021](#) for providing continuous education to children in conflict zones and refugee settings.
- **Global Societal Impact:** This "offline-first" prototype was the key technical catalyst that prompted **UNICEF** to launch the initiative using Microsoft Community Training for no-connectivity regions, contributing to a high-scale initiative that reached **7.4 million learners across 43 countries** by 2024;
- **Strategic Architectural Leadership:** Successfully advocated for, and worked on containerization and migration to **cross-platform .NET Core** in close collaboration with [Sanmoy Ray](#); this move was critical for the platform's ability to run on evolving hardware in emerging markets.
- **Operational Support & Edge Strategy:** Facilitated the integration of **Azure IoT Hub** to support remote management and observability for offline learning hubs; assisted in testing data-sync on diverse Linux-based hardware to ensure reliable curriculum delivery in remote field operations.

Swachh Bharat Mission (powered by Project Sangam)

- Directed an engineering team through the end-to-end implementation of key mobile features, culminating in the successful launch of the official [Swachh Bharat Mission eLearning mobile app](#).
- Spearheaded the rapid triage of launch-critical issues in close collaboration with [Aditya Agarwal](#) and PM teams to ensure a smooth migration of the training platform.
- Architected and implemented a high-impact search-enabled dropdown to resolve a critical UI bug; this allowed users to efficiently navigate and select from **the hundreds of cities where the platform was deployed**.
- **Contributed to a high-scale social impact project** that successfully trained and empowered over **110,000+ municipal functionaries** across **4,000+ cities** in India.

Platform Localization

- Spearheaded global accessibility by **engineering an internal localization framework** and integrating with manual translation pipelines.
- Initially developed comprehensive support for **right-to-left (RTL) formats** before assuming full platform localization ownership.
- Conducted a **comprehensive investigation on ligature support** to ensure accurate rendering of complex scripts across the ecosystem.
- **Custom PDF Certification Engine:** Architected a templated HTML-to-PDF generation feature by defining rigorous localization and security requirements; strategically collaborated with IronPDF to accelerate delivery while ensuring full support for complex scripts and global accessibility standards.

Notifications Infrastructure

- Initially onboarded via Xamarin consumption APIs before assuming full ownership of the platform's notification subsystem, under the mentorship of [Rohan Badlani](#).
- **High-Throughput Bulk Ingestion:** Architected and built an end-to-end bulk upload system enabling parallel ingestion of multi-format media; significantly reduced administrative onboarding time and **super-charged the manual testing experience** for everyone.
- **Developed a scalable monitoring infrastructure** using Event Hubs and Azure Data Explorer to centralize telemetry and enable diagnostic and analytic dashboards.
- **Automated end-to-end deployments and CI/CD pipelines** in close collaboration with [Sukhjeevan Reddy](#); enabling the Public Preview release on the Azure Marketplace.
- **Identity & Authentication:** Streamlined the SMS-based login flow by implementing custom Identity Server solutions over Azure Cosmos DB and closely collaborated with Telesign to significantly reduce SMS code-sending costs across many countries.
- **Search Infrastructure & Optimization:** Designed and implemented a custom search index featuring advanced schema optimizations and Lucene-based retrieval.
 - Used Azure Cognitive skills to extract metadata from PDFs, videos, and documents into indexed columns.
 - Improved search relevance by configuring Azure Cognitive Search with fuzzy matching, synonym maps, phonetic analyzers etc.
- **Media Security & DRM:** Authored a technical proposal to modernize content protection for temporal media using industry-standard **DRM (Digital Rights Management)** including **Widevine, PlayReady, and FairPlay** beyond just encryption-in-transit.
- **Advanced Threat Modeling:** Conducted deep-dive threat modeling and attack surface analysis using Microsoft internal security frameworks; ensuring the integrity of user data and platform content.
- **Privacy & Compliance:** Enabled automated compliance tooling to automate the identification of secure, licensed dependencies; ensuring rigorous adherence to global data protection and privacy standards.

- **State Persistence & User Analytics:** Developed features for real-time tracking of user progress, including "most recently viewed lesson" and "last watched timestamp" for temporal media content ensuring a seamless cross-device learning experience.

Microsoft Hackathon

Engineer

June — 2018

Hyderabad, Telangana, India

- **Kool AI(d) for Gamers:** Collaborated in a team of two, to apply Extracted live game metrics and game states from live DotA-2 streams for real-time analytics.
- **Koev Halev:** Actively contributed at the Microsoft IDC campus, providing first-person domain expertise in cross-functional design discussions for a project focused on health support.

Microsoft Code.Fun.Do

Engineering Lead and Pizza Logistics Moderator

2018 — 2019

India

- **Competitive Programming Strategy:** Advocated for structured programming contests to align internal technical standards with top-tier industry benchmarks.
- **Event Moderator:** Contributed to technical seminars and Pizza logistics at premier engineering institutes, including **NIT Surathkal** and **BITS Pilani (Hyderabad)**.

The Garage, Microsoft

Engineer

June — 2018

Hyderabad, Telangana, India

- **IoT Home Automation:** Collaborated with a team to retrofit legacy electric switches with motorized 3D-printed actuators and control software.

Goldman Sachs

Part-time

September 2016 — October 2016

Bengaluru, Karnataka, India

- Led a 3-member team (ft. [Samiksha Gupta](#) and [Shivam Tyagi](#)) to secure a national runners-up position in the online round of GS Quantify 2016 — A highly competitive flagship event assessing advanced interdisciplinary capabilities in algorithmic problem-solving, predictive data-science, and quantitative trading strategies; Contributed to the final Bond-Liquidity-Prediction solution.

Microsoft Foundry

Engineering Internship

May 2016 — July 2016

Hyderabad, Telangana, India

- **Conversational AI:** Engineered chat-bots using the **Microsoft Bot Framework**, architecting custom conversational flows for automated form-filling.
- **Collaborative Development:** Contributed to a high-impact project within a cross-functional team of **8 members**.
- **Computer Vision & NLP:** Helped enhance OCR performance using **OpenCV** and test refined natural language experiences using **Microsoft LUIS**.

Xerox Research Centre

Machine Learning Internship

Dec 2015 — Jan 2016

Bengaluru, Karnataka, India

- **Xerox Research Innovation Challenge:** Engineered predictive neural network models using vitals and lab data to forecast patient mortality and health complications among ICU patients.
- Secured **3rd place** (online) and **7th place** (onsite) in the challenge alongside a 3-person team, successfully competing against advanced Ph.D. research teams.
- Completed an intensive Winter School on predictive data-science under the supervision of [Vaibhav Rajan](#).

PROJECTS

- **2D Physics Game Engine:** Developed a custom rigid-body and gravitational physics engine in Java focusing on real-time kinematic interactions.
 - Implemented efficient **collision detection and resolution** for multiple dynamic objects.
 - Engineered an **environmental boundary system** to manage object interactions with surrounding walls.
 - Implemented a **Near-Earth constant gravity model** to [simulate realistic downward acceleration](#).
 - Implemented a **Newtonian $1/r^2$ gravity model** to simulate object interaction in space.
- **AIMA-CSharp Open Source Contribution:** Contributed to the [AIMA-CSharp repository](#) by implementing pseudo-code in C# from Peter Norvig's *Artificial Intelligence: A Modern Approach*.
- **Image Steganography:** Implemented a system to encrypt and embed hidden messages within images.
- **Multi-Agent Environment Simulation:** Built a 2D grid simulation of agents coordinating with one another.
- **Intel E5200 CPU Overclocking:** Successfully overclocked an Intel Pentium E5200 processor from a base clock of **2.5GHz** to **2.8GHz**. Involved tuning **FSB frequencies, CPU multipliers, and voltage settings**; Benchmarked thermal output and system stability under peak loads.
- **The Chess Game:** Developing an evolving chess engine from pre-college through my final year, transitioning from a foundation in Java to a sophisticated AI-driven system. Independently integrated an AI engine utilizing **Mini-Max** and **Alpha-Beta pruning**.
- **Tic-Tac-Toe AI:** Implemented a decision-tree based AI and multi-player for Tic-Tac-Toe.

EDUCATION

Indian Institute of Technology (Banaras Hindu University)

Integrated Master's degree (5 years): Mathematics & Computing

July 2013 - May 2018

Varanasi, Uttar Pradesh, India

- **Thesis:** Deep Learning based Classification of Handwritten Mathematical Symbols (Supported by [Tanmoy Som](#))
 - * Trained and fine-tuned transfer learning models to achieve state-of-the-art accuracy on the HasyV2 dataset.
- **Teaching Assistant** for the freshman course on **Mathematical Methods** (Mentored by [Manushi Gupta](#))
- **Problem Setter & Moderator – Hackerrank (Technex'16):**
 - * **Mathletics:** www.hackerrank.com/mathletics-technex16
 - * **Prayaas:** www.hackerrank.com/prayaas-technex16
- **Ranked 1st in CodeDef-2016**, a coding contest organized under Udyam, a college festival organized by the Electronics branch.
- Ranked 2nd in Analiticity, a campus-wide data-science contest in Technex '16.
- Ranked 1st in COPS-OPC-2015, a programming contest organized by the club of programmers at IIT Varanasi.
- **Microsoft Code.Fun.Do** Collaborated with [Sumit Sahrawat](#) in a team of 4, developing a 2D top-down space shooter game using the Unity game engine.
- **Technex-15:** Ranked 3rd in college in Mathletics, a Mathematics and Computing event.
- **Predicting Tourism Trends:** Participated in a team of 3 to build machine learning models to forecast tourism trends (Advised by [Santwana Mukhopadhyay](#))

- **Rediscovering Darkness:** Runner's up in the Astronomy Club event
- **KashiYatra-14:** Runner's up in Treasure Hunt, participated in a team of 3.
- **The Chess Game:** Stood 2nd in Technex-14 Mod-Ex; Developed an evolving chess game from pre-college, featuring a chat room etc..
- **Technex-13:** Runners-up (in a team of 5) against 100+ other teams, at the Robo-Soccer event organized by the Robotics Club; Retired-hurt while attempting to design an All-Wheel-Drive Train.
- **Event Organizer & Moderator:** Competitive tournaments for Age of Empires, Rise of Nations, Call of Duty, Need for Speed, and Defense of the Ancients!

FIITJEE

Joint Entrance Examination

April 2011 - June 2013

Dhanbad, Jharkhand, India

- **All India Rank 494** in Joint Entrance Examination — 2013, among 1.3 million candidates.
- **All India Rank 1672** in IIT-JEE (ADVANCED) — 2013, among 150k candidates.
- **State Rank 2** in Jharkhand Combined Entrance Competitive Examination (Engineering) — 2013.

Delhi Public School

All India Senior School Certificate

June 2011 - April 2013

Dhanbad, Jharkhand, India

- **Scored 96.5 %** in All India Senior School Certificate — 2013 examination!
- **All India Rank 6355** in National Eligibility cum Entrance Tests (NEET) — 2013, among 650k candidates.
- **Awarded DST INSPIRE Science Scholarship for ranking in the top 1%** of AISSCE — 2013.
- **State Rank 1** in Jharkhand Combined Entrance Competitive Examination (Medicine) — 2013.
- Recognized in the **top 0.1% of AISSCE — 2013 for excellence in Physical Education.**

De Nobili School, Sijua

Indian Certificate of Secondary Education

March 2000 - March 2011

Dhanbad, Jharkhand, India

- **Scored 87.9 %** in the Indian Certificate of Secondary Education — 2011 examination.
- Awarded Scholarship via FIITJEE Talent Reward examination.

- How The Universe Works
- Through the Wormhole with Morgan Freeman
- **MKBHD Studios:** [Dark Artistry](#), [AutoFocus](#), [Waveform](#), [Shorts](#), etc. 2013
- **Peterson Academy:** [Maps of Meaning](#), [Order & Chaos](#), [Psychology](#), [Philosophy](#) 2018
- **The AI Podcast:** [AI](#), [Blockchain](#), [Science](#), [Philosophy](#) & [Mortality](#) 2018
- **DeepLearning.AI:** [Deep Learning Specialization](#) 2018
- **Linus Media Group:** [Tech Tips](#), [Segues](#), [TechLinked](#), [The WAN Show](#), [Shorts](#), etc. 2019
- **University of Alberta & Amii:** [Reinforcement Learning Specialization](#) 2020
- **The State University of New York:** [Blockchain Basics and Smart Contracts](#) 2021
- **DeepLearning.AI:** [Generative AI For Software Development](#) 2025

ACCOMPLISHMENTS

• Competitive

- Ranked **83rd** internationally in **Google Kickstart Round D 2017**
- **ACM ICPC 2016:** Qualified for and represented IIT (BHU), Varanasi at the Asia-Amritapuri regionals
- Ranked **15th** internationally (2nd in India) in **Codeforces Round #410 (Div. 2)**
- **ACM ICPC 2015:** Qualified for and represented IIT (BHU), Varanasi at the Asia-Amritapuri regionals (ft. Shivam Tyagi and Aditya Kapoor)
- Ranked **19th nationally** in CodeChef May Challenge 2015, reaching an **overall India Rank of 31** in CodeChef's Long Challenge track.
- **IndiaHacks 2016 (ft. Sartaj Singh):** Secured a **14th rank** against 3,000+ other teams in their data-science track; Invited to the on-site finals held at **Vivanta by Taj, in Bengaluru, KA.**
- Ranked 41st in IIT Kanpur's Monthly Programming Contest in a team of 3 (ft. Abhishek Bind and Rishabh Daal) among 1117 other teams
- Ranked 16th on HackerEarth's October Sky among 800 international candidates
- Ranked 29th in IIT Bombay's TechFest – 15 International Coding Challenge Final Round
- Ranked 13th in IIT Bombay's TechFest – 15 International Coding Challenge Screening Round

SKILLS

- **Natural Languages:** English (Divine), Hindi (Ancient), Marwari (Legend), Sanskrit (Crusader), etc.
- **Leadership:** Cross-functional Collaboration, Design Reviews, Model, Coach, Care, etc.
- **Site Reliability:** Business Continuity and Disaster Recovery, On-call and Incident Response etc.
- **Engineering:** Business Continuity & Disaster Recovery, Data Structures and Algorithms, Distributed Systems and High Performance Computing (Billions of daily notifications), Data Pipelines, Authentication and Identity Systems, Blockchain, API design and Smart Contracts, etc.
- **Security:** Ethical Hacking, Threat Modeling, etc.
- **Privacy:** Encryption, GDPR, CCPA, etc.
- **Artificial Intelligence:** Reinforcement Learning (Generative Adversarial Play, Exploitation, Exploration, etc.), Deep Learning (Neural Networks, CNNs, RNNs, Transfer Learning, etc.), Machine Learning (Mathematical Modeling and Search Heuristics), Data Science (Statistics, Classification, Regression, PCA etc.), LLM Applications, etc.
- **Data & Storage:** Redis, NoSQL, SQL, Blockchain, etc.
- **Search:** Apache Lucene, Elasticsearch etc.
- **Cloud & Infrastructure:** Microsoft Azure (Expert), AWS, Kubernetes, CDN, CI/CD & Infrastructure Automation, etc.
- **Frameworks & Libraries:** JVM, .NET, IronPDF, React, TensorFlow, Keras, Pandas, Scikit-learn, XGBoost, OpenCV, Unity, etc.
- **Programming Languages:** Logo, Basic, C, C++, Java, Octave, Python, C#, PowerShell, Hack, etc.
- **Philosophy** — Ancient

PURSUIITS

- Metaverse — MMORPG, RTS, FPS, Podcasts, Documentaries, Anime, Books, WIP.
- Science — Allopathy, Anatomy, Silicon Hardware, Theoretical Physics, WIP.
- Visual Awakening — Psychedelics etc.
- Music — Pop and Alternative Rock, Lo-Fi, EDM, MeaningWave, Rap, etc.
- Guitar — Beginner (Electric, Yousician, WIP)
- Cooking — Beginner (Taste (Waikiki, BHU[Vishwanath Temple], eat.fit, Jack in the Box, etc.), WIP)
- Swimming — Beginner.
- Traveling — Stars, NASA, SpaceX, Amazon Leo, Auroras, Universities, Museums, Monuments, etc.
- Comedy — Bill Maher, Stephen Colbert, Hasan Minhaj, Trevor Noah, Joe Rogan, Dave Chappelle, etc.
- Languages — Mandarin, Punjabi, Korean, Spanish, Tamil, Japanese, etc.

PERSONAL

- Sangeeta Jiloka (Artist, Physics, Cooking, Coach, Allopathy)
- Raj Kumar Agarwal (Reporter, Journalist, Lead Editor, Business, Law)
- [Parul Aggarwal](#) (Experimental Physicist)